

Nijbhasha

निजभाषा

A Hindi-medium teacher teaches in Santali. Offline, on a tablet.

Children are taught in a language many do not speak at home

Jharkhand, Grade 1: the teacher speaks Hindi; the child may speak Santali, Ho or Mundari.

~98%

of surveyed schools teach in Hindi

17 · 13 · 7 %

Ho · Santali · Mundari: Grade 1 home languages

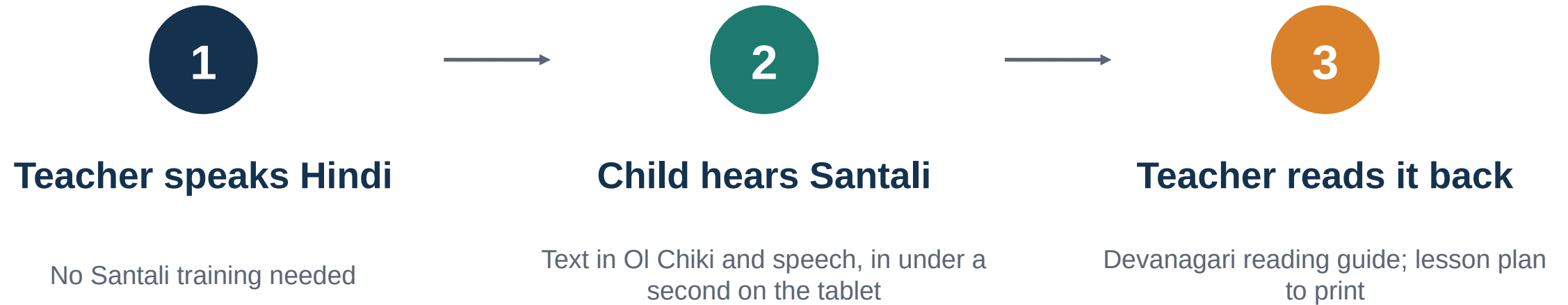
5,000+

tribal-area primary schools named in the problem statement

PALASH shows mother-tongue teaching works; it cannot find enough teachers who speak these languages.

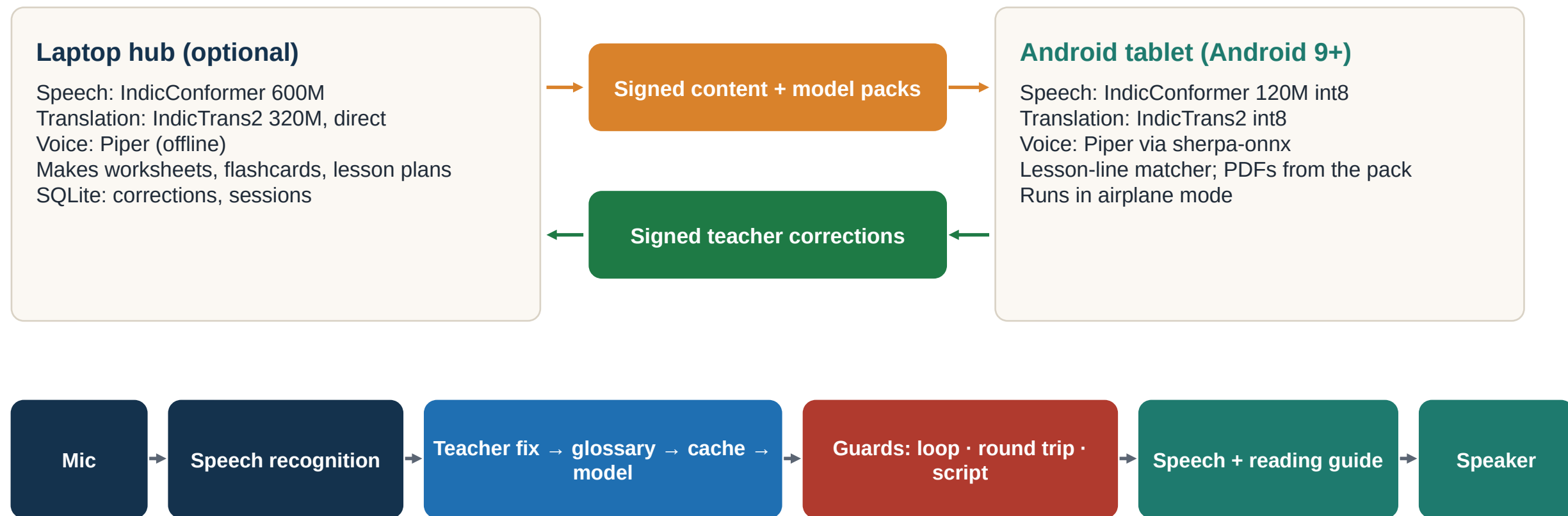
A child who cannot answer “दो और तीन कितने होते हैं?” may not be failing at addition, but at Hindi.

Not a translator: a bridge the teacher can use tomorrow



The community corrects it; every doubtful line is flagged, never played as right.

One pipeline, two devices, no internet after the first sync



The same page (frontend.html) runs on both; the same 24-case REST contract is tested on both.

R1: every lesson line in Santali text and speech, with a trust tier

- Human first: teacher correction → verified sentence → cache → model
- Direct Hindi → Santali (IndicTrans2), Ol Chiki script
- Speech offline; Devanagari reading guide for the teacher
- Tier A human-written · B model · C flagged



Live page: Santali (Ol Chiki), reading guide in Devanagari, badge "B · Model".

Santali is spoken by a Hindi voice reading our transliteration: no offline Santali voice exists; two alternatives lost on a pre-set rule.

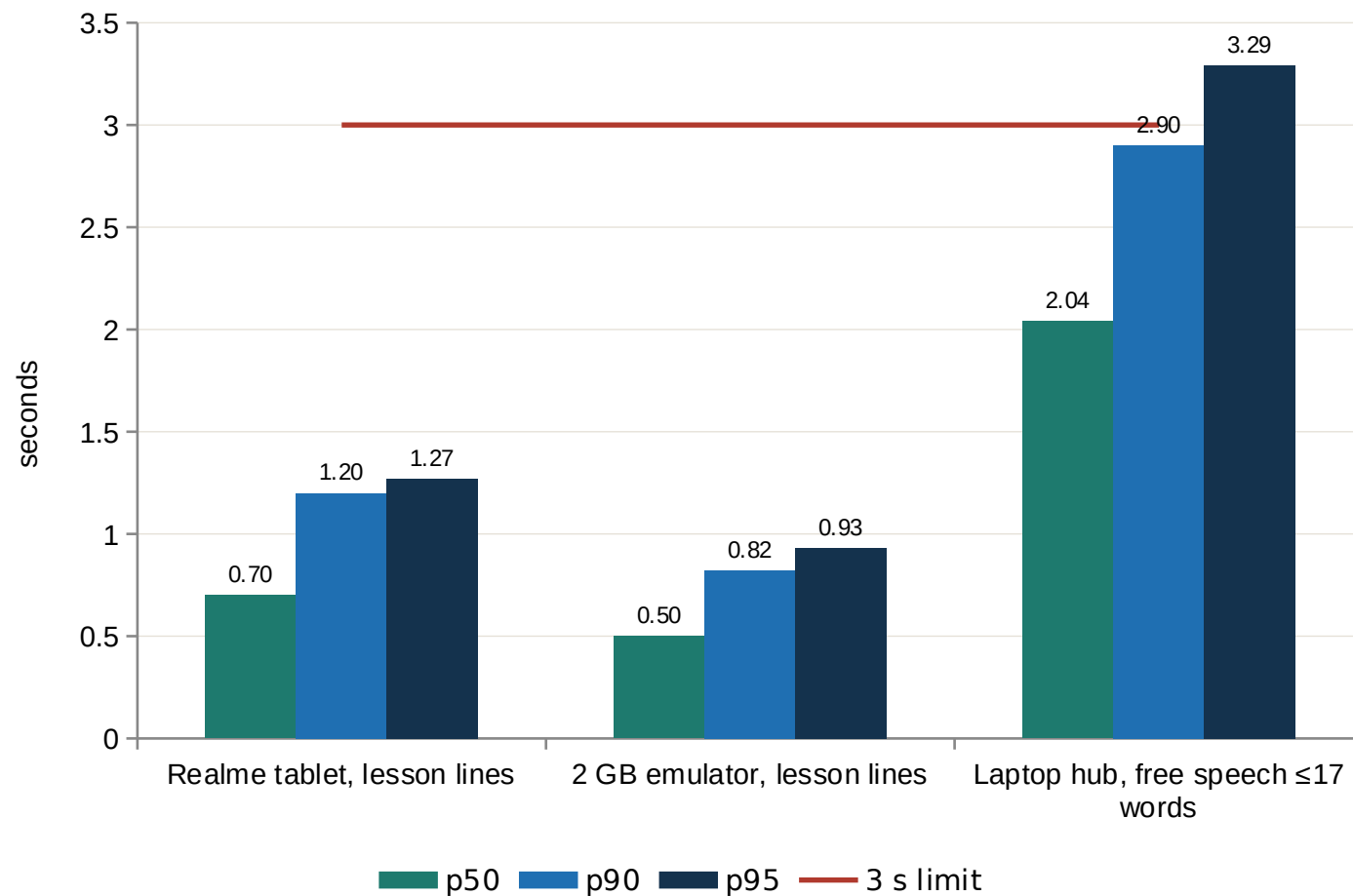
32.2

chrF++ Hindi → Santali,
IN22-Conv (n = 1,503)

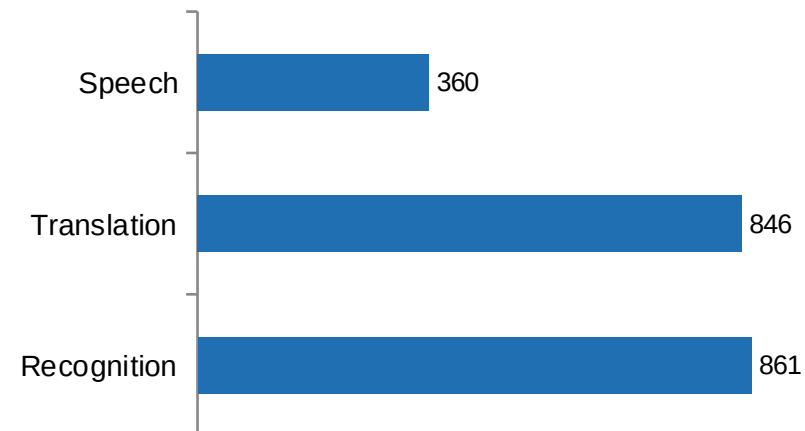
119

lesson lines checked: script,
numbers, flags

R2: spoken lesson lines on the tablet take 0.70 s (median), 1.27 s at p95



Laptop hub, per stage (median)



Honest tail: on the laptop, 8 of 96 free-speech sentences took over 3 s (p95 3.29 s). Free speech stays on the laptop; the tablet matches lesson lines and refuses anything else.

R3: each lesson prints a worksheet, flashcards and a teacher plan, tagged to NIPUN

नजिभाषा — अभ्यास पत्रक / bṛ.ḍṛ bṛḍṛḍṛ

Simple Addition | कक्षा 2 / bṛḍṛ 2 | 28.09.2026

NIPUN संस्था / NIPUN Pṛḍṛḍṛ: NIPUN-G1-NUM-2

नाम: _____ तारीख: _____

1. गिने और लिखो

























R4: airplane mode, 24 of 24 checks on a 2 GB Android 9 emulator and a real tablet

Android 9+

minSdk 28

72 MB

release APK

361 MB

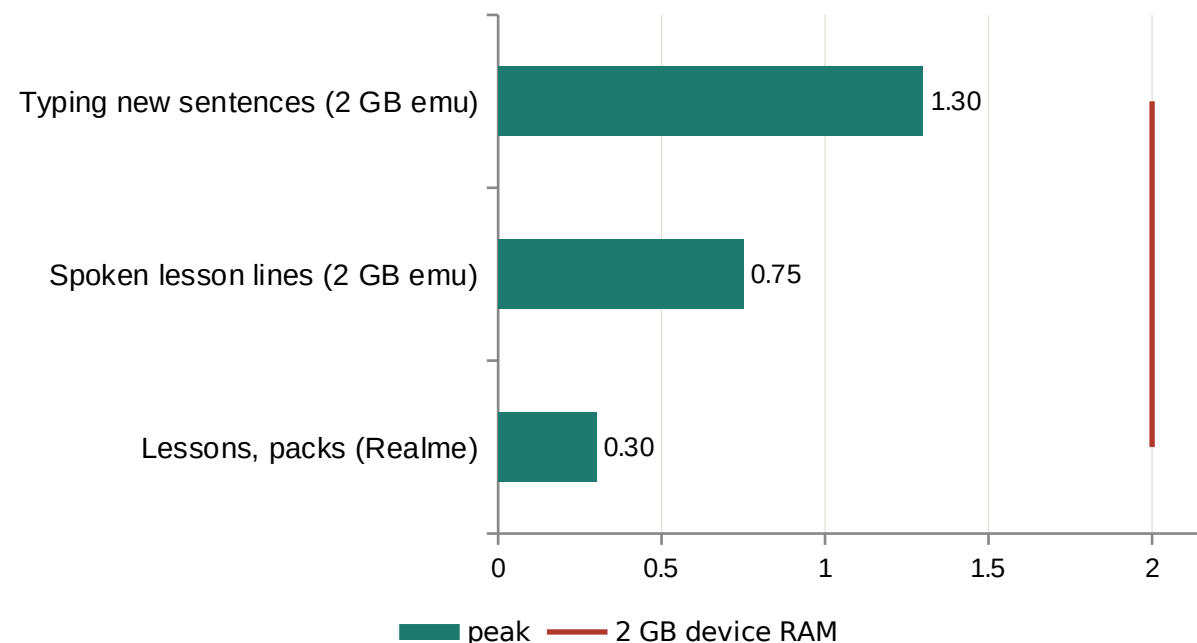
model pack, int8, signed

38 MB

content pack, signed

Tested: Realme Pad Mini (4 GB, Android 11) and a 2 GB Android 9 emulator. Not yet: a real 2 GB tablet, a 32-bit build.

Peak memory (app + WebView), GB



Translation (1.3 GB) loads only when needed; the recogniser is released first.

Eight things a classroom needs that a translator does not do

built

Teacher corrections

reused first; synced between tablets as signed files

new

Devanagari reading guide

the teacher can read the Santali aloud

built

Trust tiers + 3 guards

loop, round trip, script; C is never auto-played

sample

Local-context lessons

teachers import village examples; app translates

new

Lesson plans

daily script with pronunciation guide and answers

built

Progress per NIPUN Lakshya

by week; CSV and PDF; no names

new

Community voice corpus

adults only, consent, stays on the laptop

built

Low-resource bootstrapping

glossary-first; Mundari by LoRA (preview)

No native speaker yet: so nothing doubtful is played as if it were right



- Round-trip check: a caution, not a score
- Script guard: catches words in other scripts
- Every Santali line on a native-review list
- No cloud, no names; reading audio deleted

A flagged line: tier C, a warning, no auto-play, the nearest verified sentence offered.

13 of 119

lesson lines with Urdu / Meetei
letters: now tier C

0.46 / 0.45

round-trip flag precision / recall

Where we win: the classroom job, not the translation model

	Nijbhasha	Translate wrapper	Online-only app	Text-only tool	LLM chatbot
Works with no internet	Yes, tablet in airplane mode	No	No	Sometimes	No
Voice to voice	Both ways; 0.70 s on tablet	Some apps	Network-bound	No	Slow, online
Santali in Ol Chiki	Yes; Mundari preview	Santali only	Varies	Varies	Unreliable
NIPUN lessons + worksheets	18 lessons, every Lakshya	No	Rarely	Sometimes	Ad hoc
Flags doubtful output	Tiers + 3 guards	No	Rarely	Rarely	No
Child data stays local	Yes	Cloud	Cloud	—	Cloud

We do not claim better translation quality than any of these.

Built for PALASH scale: 1,041 schools today, 5,000+ in the brief

1,041

PALASH schools, 8 of 24 districts

5,000+

tribal-area schools in the brief

1 tablet

per classroom; laptop optional

₹0

licence or per-use fees

How it grows

- Santali now → Mundari (our LoRA, preview) → Ho once text exists
- Same pipeline for other Munda and low-resource languages
- Content = signed packs a district can build and send
- Proposed first step: a PALASH classroom pilot (not yet agreed)

Assumptions: hardware and training costs are not estimated; school counts are PALASH's and the brief's, not our reach.

Roadmap: close the known gaps first, then grow with the community

30 days

- Native review of the 18 lessons
- 32-bit ARM build; a real 2 GB tablet
- APK release on GitHub

90 days

- Proposed PALASH pilot, 5 classrooms
- Children's speech benchmark
- Mundari on the tablet

180 days

- Ho content with native writers
- Voice from the community corpus
- Resumable pack download

Sustained by: open models and MIT code; lessons as editable JSON; corrections and the voice corpus grow with use.

See it run: airplane mode, the timer on screen

- 1 Tablet in airplane mode: lessons load
- 2 Teacher says a lesson line → Santali in 0.7 s
- 3 Reading guide under the Santali
- 4 A doubtful line: tier C, not played
- 5 Worksheet, flashcards, lesson plan print



github.com/barath0512s-rgb/nijbhasha

Demo video: [add link after recording]

Our ask: a native Santali reviewer and one PALASH classroom.

A1 · Data sources and licences

What	Used for	Licence
AI4Bharat IndicConformer 600M / 120M	Speech recognition (hub / tablet)	MIT
AI4Bharat IndicTrans2 indic-indic 320M	Translation, hub fp32 and tablet int8	MIT
Piper hi_IN-pratham-medium	Speech (Hindi, and Santali via transliteration)	CC BY-NC-SA 4.0
sherpa-onnx 1.13.8 (with espeak-ng)	On-device speech	Apache-2.0 (espeak-ng GPL-3.0: APK under GPL-3.0)
Meta MMS voices mms-tts-unr / -hoc	Mundari and Ho voices (preview)	CC BY-NC 4.0
MMLoSo 2025 Hindi-Mundari data	Mundari LoRA (18,978 training pairs)	CC BY-SA 4.0
IN22-Gen, IN22-Conv; FLORES-200	Evaluation only (leakage-guarded)	CC BY 4.0; CC BY-SA 4.0
FLEURS (Hindi), IndicVoices (Santali valid.)	Speech benchmarks, adult	CC BY 4.0
OpenMoji; Noto fonts	Pictures; Devanagari and Ol Chiki type	CC BY-SA 4.0; OFL
Our code	Everything else	MIT

A2 · Benchmarks (all offline)

Measure	Setting	Result
Voice to voice, lesson lines	Realme Pad Mini, n = 50	p50 0.70 · p90 1.20 · p95 1.27 s
Voice to voice, lesson lines	2 GB Android 9 emulator, n = 50	p50 0.50 · p90 0.82 · p95 0.93 s
Hindi → Santali, free speech ≤ 17 words	Laptop hub, 3 runs × 32 (FLEURS)	p50 2.04 · p90 2.90 · p95 3.29 s; 8 of 96 > 3 s
Santali → Hindi, upload to reply	Laptop hub, 80 answers per run	p90 2.52–2.60 s
Typed new sentence on the tablet	Realme / 2 GB emulator	p50 1.50 s / 0.60 s
Translation quality hi → sat	IN22-Gen / IN22-Conv / FLORES-200	chrF++ 31.3 / 32.2 / 27.4
Mundari preview (our LoRA)	Held-out 5 % of MMLoSo, n = 1,021	chrF++ hi → unr 30.92 · unr → hi 35.97
Speech recognition, WER (normalised)	Hub 600M: Hindi / Santali	12.5 % / 31.0 %
Peak memory	Lessons / lesson-line voice / typing	0.30 / 0.75 / 1.30 GB

A3 · How we keep the numbers honest

- 336 automated tests pass without models; 40 more need the models; CI on every push
- A test fails if the README quotes a number with no evidence file
- Model choices by rules written before the results (voice, decoding, int8)
- Leakage guard: no benchmark sentence can reach training
- Same 24-case REST contract tested on the hub and the tablet
- Every figure printed by a script: `tools/deck_numbers.py`, `tools/latency_percentiles.py`

A4 · Security and privacy

- No cloud at run time; network sockets blocked in the offline test
- Packs signed (Ed25519) and verified on the tablet; a changed pack is refused
- Progress reports: no names, no audio
- Reading check: consent box; the recording is deleted after scoring
- Voice corpus: adults only, consent, no names; export only from the laptop, only with consent to share
- Hub over HTTPS with its own local certificate for tablet microphones

A5 · Limitations we disclose

- No native-speaker review yet of Santali, Mundari or Ho output, or of the voice
- Santali voice is a Hindi voice reading a transliteration
- No real 2 GB tablet; no 32-bit ARM build
- Free-form speech on the laptop only; p95 3.29 s there
- Children's speech and classroom noise not measured
- Ho: voice only; Mundari: preview on the laptop
- NIPUN mapping not reviewed by a teacher

A6 · Real lesson lines, as the app shows them

Hub output from the content pack (26 Sep); not reviewed by a native speaker.

Hindi (teacher)	Santali, Ol Chiki (child hears)	Reading guide, Devanagari	Tier
अपनी उंगलियां दिखाओ और मेरे साथ गिनो।	ᱡᱤᱨᱤ ᱦᱚᱛᱟᱜ ᱵᱤᱨᱫᱟ ᱥᱚᱝ ᱠᱚ ᱡᱤᱨᱤ ᱮᱞᱴ ᱢᱚᱰᱟᱲ ᱦᱚᱱᱚᱛ ।	आमाक् उंग्लि देखाव् मे आर् अमाक् सां गिन्ति मे।	A glossary
यहाँ कितने पत्थर हैं? बताओ।	ᱦᱚᱱᱚᱛ ᱦᱚᱛᱟᱜ ᱪᱤᱨᱫᱟ ᱦᱚᱱᱚᱛ-ᱠᱚ? ᱯᱟᱨᱴ ᱦᱚᱱᱚᱛ ।	नेले जौतों धिरि मेनाक्आ? लय् मे।	A glossary
तीन और चार कितने होते हैं?	ᱦᱚᱱᱚ ᱠᱚ ᱦᱚᱱᱚ ᱦᱚᱛᱟᱜ ᱣᱟᱹᱨᱩᱛ-ᱠᱚ?	पे आर् पुन् जौतों हुयुक्आ?	A glossary
आठ में से पांच घटाओ। उत्तर क्या है?	ᱚᱸᱰᱚ.ᱮ ᱪᱤᱨᱫᱟ ᱦᱚᱱᱚᱛ ᱪᱚᱱ ᱦᱚᱱᱚᱛ । ᱢᱚᱰᱚ ᱦᱚᱱᱚᱛ ᱪᱚᱱᱚᱛ?	इरल् खॉन् मॉणे कॉम् मे। तेला चेत् काना?	A glossary
चार समूह बनाओ, हर समूह में दो पत्थर रखो।	ᱦᱚᱱᱚ ᱢᱚᱱᱚᱛ ᱢᱚᱱᱚᱛ ᱦᱚᱱᱚᱛ ᱦᱚᱱᱚᱛ, ᱦᱚᱛᱟᱜ ᱢᱚᱱᱚᱛ ᱚᱱ ᱦᱚᱱᱚᱛ ᱪᱤᱨᱫᱟ ᱪᱚᱱᱚᱛ ᱦᱚᱱᱚᱛ ।	पोन् गौटां गादेल् बेनाव् मे, जौतों गादेल् रे बार्या धिरि दौहों मे।	B model
सबसे छोटी चीज़ पहले रखो।	ᱦᱚᱱᱚᱛ ᱚᱱ ᱣᱟᱹᱨᱩᱛ ᱚᱱᱚᱱᱚ ᱪᱚ ᱪᱚᱱᱚᱛ ᱦᱚᱱᱚᱛ ।	पोय्लो रे हुडिज् जिनिस् कॉ दौहों मे।	C model · round-trip
आज रविवार है।	ᱢᱚᱱᱚᱛ ᱪᱚ انوار ا ۔	तेहेज् दाँ اتوار ا۔	C cached · script guard

A7 · Questions we expect

Question	Short answer
Is the Santali correct?	Not measured without native speakers; human-written first, doubtful lines flagged, review list ready
Does it run on 2 GB?	2 GB Android 9 emulator yes; real 2 GB tablet not yet; typing needs 1.3 GB, loaded on demand
Under 3 s?	Tablet lesson lines p95 1.27 s; laptop free speech p95 3.29 s (8 of 96 over)
Children's voices?	Not measured; first pilot benchmark
Ho and Mundari?	Mundari translation preview (our LoRA); Ho voice only, no licensed text
Why not an LLM?	Needs internet or far more memory; can invent
Cost?	No licence or per-use fees; hardware not estimated
Privacy?	No cloud; no names; reading audio deleted; corpus adults-only with consent